

*11. Math
II
Algebra*

Detail Course:

1. System of Linear Equations:

Linear equation; definition, examples

System of linear equation; consistent, inconsistent, unique solution, no solution, infinitely many solutions and their examples

General Systems of linear equations; coefficient matrix, the vector of unknowns, the right hand-side vector, and the augmented matrix

Gaussian elimination; definition and examples

Elementary replacement and scale operations; definition and example

Row equivalent pairs of matrices; definition of matrix, pivot row, row operation, row equivalent, equivalent matrices, and their examples, equivalence relation

Elementary row operations; definition, scaling of an equation, replacement, scale, swap

Reduced row echelon form; definition, pivot, and their examples

Theorem : Every matrix has one and only one reduced row echelon form

Pivot position and examples

Row echelon form; definition, difference between reduced and row echelon form

Intuitive interpretation

Application: Feeding bacteria with an example

General exercises 1.1 (no. 1-29, 31, 40-45, 47, 48, 49, 50)

2. Vectors and Matrices:

vectors; definition and examples

Linear combinations of vectors; definition, examples

Matrix- vector products; definition and examples

The span of a set of vectors; definition, examples

Interpreting linear systems; discussion by an example

Row equivalent systems,

Theorem: No proof : Let two linear systems of equations be represented by their augmented matrices. If these two augmented matrices are row equivalent to each other, then the solutions of the two systems are identical

Row equivalent, examples

Consistent and inconsistent systems;

Theorem : A system of linear equations, $Ax = b$ is consistent if and only if the vector b is in the span of the set of columns of A .

Theorem: No proof : Let A be an $m \times n$ matrix. The system of equations $Ax = b$ is consistent for all b in $\mathbb{R}^m$ if and only if the columns of A span $\mathbb{R}^m$. In other words, $\text{Col}(A) = \mathbb{R}^m$

Theorem : Let A be an $m \times n$ matrix. The system of equations $Ax = b$ is consistent for all b in $\mathbb{R}^m$ if and only if each row of the coefficient matrix A has a pivot position.

Theorem : A system of linear equations is consistent if and only if its augmented matrix has a pivot position in the last column.

Theorem: No proof : A system of linear equations is consistent if and only if the reduced row echelon form of its augmented matrix does not have a pivot position in the last column.

General exercises 1.2 (no. 1- 14, 1, 19, 20, 24, 26, 30, 31, 32, 33, 34, 35, 38, 40)

Kernel or null spaces of a matrix; definition and examples

Theorem : If x and y are vectors in $\text{Ker}(A)$, and if α is a scalar, then $x+y$ and αx are also in $\text{Ker}(A)$.

Theorem : If $Ax = b$ and $y \in \text{Ker}(A)$, then $A(x+y) = b$.

Theorem : If $Ax = Ay$, then $x - y \in \text{Ker}(A)$

Theorem: No proof : If u is a vector such that $Au = b$, then every solution of the equation $Ax = b$ is of the form $x = u + z$, for some vector z in $\text{Ker}(A)$

Theorem : If two matrices are row equivalent to each other, then their kernels are the same.

Homogeneous equations; definition and examples

Uniqueness of the reduced row echelon form;

Theorem : Every matrix has one and only one reduced row echelon form

Rank of a matrix; definition and examples

Corollary: No proof: The rank of a matrix is the number of pivots in its reduced row echelon form, which is the same as the number of pivot positions in the matrix.

example

Theorem : The rank of an $m \times n$ matrix cannot be greater than n or m . In symbols, we have $\text{Rank}(A) \leq \min(m, n)$.

Theorem: No proof : These properties of an $m \times n$ matrix A are equivalent to each other.

(a) The rank of A is less than n

(b) The reduced row echelon form of A has fewer than n nonzero rows.

(c) The matrix A has fewer than n pivot positions.

(d) At least one column in A has no pivot positions.

(e) There is at least one free variable in the system of equations $Ax = 0$

(f) The system $Ax = 0$ has some nontrivial solutions.

Corollary: No proof: A homogeneous system of linear equations in which there are more variables than equations must have some nontrivial solutions

General solution of a system; example

Matrix- matrix product ; definition and examples

Indexed sets of vectors: Linearly dependence and independence; definition, examples

Theorem : If an indexed set of two or more vectors is linearly dependent, then some vector in the set is a linear combination of the others.

Theorem : If an indexed set of two or more vectors in $\mathbb{R}^m$ is linearly dependent, then some vector in the list is a nontrivial linear combination of vectors preceding it in the list.

Theorem : The rows of a matrix form a linearly dependent set if and only if there is a zero row in any row echelon form of that matrix,

Using the row reduction process; example

Theorem: No proof : The column vectors of a matrix form a linearly dependent set if and only if there is a column having no pivot.

Theorem: No proof : The column vectors of a matrix form a linearly independent set if and only if there is a pivot position in each column of the matrix.

Example

Determining linear dependence or independence; discussion

Application: Linear ordinary differential equations with an example

General Exercises 1.3 (no. 1-18, 22, 30, 31, 33, 37, 39, 40, 43, 53, 57, 58, 59, 60, 62, 63).

3. Vector Spaces:

n-Tuples and vectors; definition and examples

vector addition and multiplication by scalars; definition and examples

Properties of $\mathbb{R}^n$ as a vector space; definition, basic properties of the vector spaces $\mathbb{R}^n$

Linear combinations; definition and examples

span of a set of vectors; definition and examples

Geometric interpretation of vectors

General exercises 2.1 (no. 1- 20, 29-31)

Line passing through origin

Lines in $\mathbb{R}^2$; discussion, translation, parametric line, and examples

Lines in $\mathbb{R}^3$; discussion and examples

Theorem : Let L_1 and L_2 be two lines in $\mathbb{R}^n$ describe as

$$L_1 = \{\mathbf{u} + t\mathbf{v} : t \in \mathbb{R}\}$$

$L_2 = \{\mathbf{w} + s\mathbf{z} : s \in \mathbb{R}\}$. These lines are the same if and only if $\mathbf{u}-\mathbf{w}$ and $\mathbf{v}$ are multiples of $\mathbf{z}$,

examples

Lines and planes in $\mathbb{R}^n$; discussion and examples

General solution of a system of equations; discussion and examples

Applications: elementary mechanics, network problems, traffic flow with examples.

General exercises 2.2 (no. 1-22)

4. Linear Transformation:

Functions, mappings, and transformations; definition, examples

Domain, co-domain, and range; definition and various examples

Injective and surjective mappings; definition and examples, bijective mapping, and examples

Linear transformations; definition and examples

Theorem : Let $\mathbf{A}$ be an $m \times n$ matrix. The mapping $\mathbf{x} \mapsto \mathbf{Ax}$ is linear from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Examples

Theorem : Let T be a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$. Then there is an $m \times n$ matrix $\mathbf{A}$ such that $T(\mathbf{x}) = \mathbf{Ax}$ for all $\mathbf{x}$ in $\mathbb{R}^n$.

Example

Theorem : A linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ is completely determined by the images of the standard unit vectors in $\mathbb{R}^n$, which we have written as $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$.

Example

Injective and surjective linear transformations;

Theorem : Let $\mathbf{A}$ be an $m \times n$ matrix. For the linear map $\mathbf{x} \mapsto \mathbf{Ax}$ to be surjective (onto), it is necessary and sufficient that the columns of $\mathbf{A}$ span $\mathbb{R}^m$.

Definition of kernel and its example

Theorem : In order that the linear map $\mathbf{x} \mapsto \mathbf{Ax}$ be injective (one-to-one), it is necessary and sufficient that the kernel of $\mathbf{A}$ contain only the $\mathbf{0}$ -vector.

Effects of linear transformations; definition and examples, rotation, orthogonal projection, shear, reflection

Effects of transformations on geometrical figures;

Theorem : A linear transformation maps one line segment into another.

Theorem : In $\mathbb{R}^2$, a counterclockwise rotation of every point by an angle ϕ is a linear transformation whose matrix is

$$\begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}.$$

Composition of two linear mappings; definition and example

Theorem : If S and T are linear maps obeying the preceding equation, then SoT is linear. That is, composition of two linear maps is also linear.

Theorem : Let $\mathbf{A}$ be an $m \times n$ matrix, and let $\mathbf{B}$ be an $n \times k$ matrix. Define $S(\mathbf{x}) = \mathbf{Ax}$ and $T(\mathbf{y}) = \mathbf{By}$. Then $(SoT)(\mathbf{x}) = (\mathbf{AB})\mathbf{x}$.

General exercises 2.3 (no. 1-32, 34, 37, 43)

Vector spaces; definition and examples

✓ **Theorem** : In any vector space, if c is a scalar, then $c\mathbf{0} = \mathbf{0}$.

✓ **Theorem** : In a vector space, if $\mathbf{x}$ is a vector and c is a scalar such that $c\mathbf{x} = \mathbf{0}$, then either $c = 0$ or $\mathbf{x} = \mathbf{0}$.

✓ **Theorem** : In a vector space, for each $\mathbf{x}$, the point $\tilde{\mathbf{x}}$ is uniquely determined.

✓ **Theorem** : In a vector space, every vector $\mathbf{x}$ satisfies the equation $0\mathbf{x} = \mathbf{0}$.

✓ **Theorem** : In any vector space, we have $(-1)\mathbf{x} = \tilde{\mathbf{x}}$.

Various examples

Linearly dependent sets; definition and examples

✓ **Theorem** : Each vector in the span of a linearly independent set (in a vector space) has a unique representation as a linear combination of elements of that set.

examples

• **Theorem: No proof** : Any (indexed) set of more than n vectors in $\mathbb{R}^n$ is necessarily linearly dependent.

Theorem: No proof : Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ be a linearly dependent indexed set of at least two vectors (in a vector space). Then some vector in the indexed list is a linear combination of preceding vectors in that list.

Theorem : If a vector space is spanned by some set of n vectors, then every set of more than n vectors in that space must be linearly dependent.

Theorem : Let S be a linearly independent set in a vector space V . If $\mathbf{x} \in V$ and $\mathbf{x} \notin \text{Span}(S)$, then $S \cup \{\mathbf{x}\}$ is linearly independent.

Example

Linear mapping; discussion and examples

Theorem : Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a linearly independent set in some vector space. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be arbitrary vectors in another vector space. Then there is a linear transformation T such that $T(\mathbf{u}_i) = \mathbf{v}_i$ for $1 \leq i \leq n$.

Theorem : If T is a linear transformation, then any linear dependence of the type $\sum_{i=1}^n a_i \mathbf{u}_i = \mathbf{0}$ must imply $\sum_{i=1}^n a_i T(\mathbf{u}_i) = \mathbf{0}$.

Application: Models in economic theory; Leontief closed model.

5. Matrix Operations:

Matrix addition and scalar multiplication; definition, examples and conformal to addition and scalar multiplication

Matrix-matrix multiplication; definition, example and conformal to multiplication

Pre-multiplication and post-multiplication of matrix; definition and example

Theorem: No proof : The (i,j) -element in the product of two matrices $\mathbf{A}$ and $\mathbf{B}$ can be written in summation notation or the dot product of row i in $\mathbf{A}$ with column j in $\mathbf{B}$:

$$(\mathbf{AB})_{ij} = \sum_{r=1}^n a_{ir} b_{rj} = \mathbf{r}_i \cdot \mathbf{b}_j.$$

Example

Special matrices; definition of diagonal matrix, Kronecker delta, upper and lower triangular matrix, examples

Theorem : For any $n \times n$ matrix $\mathbf{A}$, we have $\mathbf{AI}_n = \mathbf{I}_n \mathbf{A} = \mathbf{A}$.

Theorem: No proof : The set of upper triangular matrices in $\mathbb{R}^{n \times n}$ is a subspace of $\mathbb{R}^{n \times n}$. That is, it is a vector space in its own right.

Example

Theorem : The product of two upper triangular matrices (of the same size) is also upper triangular.

Matrix transpose; definition and example

Theorem : If the product $\mathbf{AB}$ of two matrices $\mathbf{A}$ and $\mathbf{B}$ exists, then $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$.

Theorem:

(a) $(A^T)^T = A$.

(b) $(A + B)^T = A^T + B^T$

(c) $(cA)^T = cA^T$.

Symmetric matrices; definition and example

Skew-symmetric matrices; definition and examples, relation between them

Non-commutativity of matrix multiplication; example

$AB \neq BA$ by examples

Associativity law for matrix multiplication

Theorem: Matrix multiplication is associative. Thus, if the product exist, we have

$(AB)C = A(BC)$ and $A(B+C) = AB+AC$.

Trace of a matrix and its properties

Linear transformations;

Theorem: Let S and T be two linear transformations defined by $S(y) = Ay$ and $T(x) = Bx$. (A and B are matrices). If the composition SoT exists, then its matrix is AB .

Elementary matrices;

Theorem: No proof: If the matrix $\tilde{A}$ is the result of applying a row operation R to the matrix A , and if E is the matrix that results from applying R to I , then $\tilde{A} = EA$

More on the matrix-matrix product; discussion and example

Vector-matrix product; discussion and example

Theorem: No proof: If y^T is a row vector, then $y^T A$ is a linear combination of the rows of A , with coefficients taken to the components of the row vector y^T :

$y^T A = [y_1, y_2, \dots, y_n] A = y_1 r_1 + y_2 r_2 + \dots + y_n r_n$ where r_i are the rows of A .

Theorem: If the rows of A are denoted by $r_1, r_2, \dots, r_m$, then $AB =$

$$\begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix} B = \begin{bmatrix} r_1 B \\ r_2 B \\ \vdots \\ r_m B \end{bmatrix}$$

Application: diet problem, discussion and examples

General exercises 3.1 (no. 1-13, 15, 16, 17, 18, 19, 33, 43, 44, 48, 57, 59

Solving systems with a left inverse;

Theorem: No proof : If the system of equation $Ax = b$ is consistent, and if a matrix B exists such that $BA = I$, then the system of equations has a unique solution, namely $x = Bb$.

Examples

Solving systems with a right inverse; definition

Theorem: No proof : Suppose that two matrices A and B have the property that $AB = I$. Then the system of linear equations $Ax = b$ has at least one solution, namely $x = Bb$.

Example

Analysis; definition of right and left inverse with examples

Theorem: No proof : If the matrix A has a right inverse, then for each b the equation $Ax = b$ has at least one solution. If A has a left inverse, then that equation has at most one solution.

Theorem : If A and B are square matrices such that $BA = I$, then $AB = I$.

Theorem : If A , B , and C are square matrices such that $AB = AC = I$, then $B = C$

Invertible matrices; definition of invertible matrix, singular and non-singular matrix.

Theorem: No proof : If A is invertible, then for any prescribed b the equation $Ax = b$ has one and only one solution, namely $x = A^{-1}b$.

Theorem : If A , B and C such that $AC = BA = I$, then $B = C$

Examples

Theorem : For invertible $n \times n$ matrices A and B , we have $(AB)^{-1} = B^{-1}A^{-1}$. Thus, the product of two invertible matrices is invertible. The result extends to the product of any number of invertible matrices.

Elementary matrices and LU factorization; discussion and examples

Computing an inverse; discussion and example

More on left and right inverses of non-square matrices; examples

Invertible matrix theorem; discussion and **Theorems: No proof.**

General exercises 3.2 (no. 1-4, 7- 21, 27, 30, 31, 33, 35, 40, 43, 52)

6. Determinants:

Properties of determinant; definition of determinant, notation, difference between matrix and determinant, properties, examples

Theorem : If the matrix $\tilde{A}$ is obtained from the matrix A by swapping a pair of rows, then $\text{Det}(\tilde{A}) = -\text{Det}(A)$.

Example

Theorem: No Proof : There is a one and only one determinant function having properties I, II, and III.

Examples

Theorem : The determinant of a 2×2 matrix follows this rule:

$$\text{Det} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc .$$

Examples

Algorithm without scaling; discussion and examples

Zero determinant; discussion

Theorem: No proof : The determinant of a matrix is 0 if and only if the rows (or columns) of the matrix form a linearly dependent set.

Example

Theorem : A square matrix is invertible (non-singular) if and only if its determinant is nonzero.

Examples

Calculating areas and volumes; discussion

Theorem : In $\mathbb{R}^2$, if a triangle has vertices 0 , u , and v , then area of the triangle is one-half of the absolute value of the 2×2 determinant having rows (or columns) u and v :

$$\text{Area} [\Delta(0, u, v)] = \frac{1}{2} |\text{Det} \begin{bmatrix} u_1 & u_2 \\ v_1 & v_2 \end{bmatrix}|.$$

Examples

Theorem : The area of a parallelogram generated by the three points 0 , u , and v in $\mathbb{R}^2$ is the absolute value of the determinant having rows (or columns) u and v :

$$\text{Area} (0, u, v, u+v) = \left| \text{Det} \begin{bmatrix} u_1 & u_2 \\ v_1 & v_2 \end{bmatrix} \right|.$$

Theorem : Let S be a triangle or parallelogram in $\mathbb{R}^2$ having vertex at 0 . Let T be a linear mapping from $\mathbb{R}^2$ to $\mathbb{R}^3$, given by matrix A , so that $T(x) = Ax$. Then $\text{Area} [T(S)] = |\text{Det}(A)| \cdot \text{Area} (S)$.

Example

General exercises 4.1 (no. 1-22)

Minors and cofactors; definition and examples

Theorem: No proof: Let A an $n \times n$ matrix whose minor are denoted by M^i . Then these formulas are valid:

$$\text{Det}(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \text{Det}(M^{ij}) \text{ (expansion using row } i)$$

$$\text{Det}(A) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \text{Det}(M^{ij}) \text{ (expansion using column } j)$$

Examples

Definition of cofactor expansion

Theorem: No proof: Let A be an $n \times n$ matrix whose cofactors are denoted by C_{ij} . Then these formulas are valid:

$$\text{Det}(A) = \sum_{j=1}^n a_{ij} C_{ij} \text{ (expansion using row } i)$$

$$\text{Det}(A) = \sum_{i=1}^n a_{ij} C_{ij} \text{ (expansion using column } j).$$

Example

Direct method for computing determinants; discussion

Properties of determinants

Lemma: If E is an $n \times n$ elementary matrix and if A is any $n \times n$, then

$$\text{Det}(EA) = \text{Det}(E) \text{Det}(A)$$

Theorem: For square matrices of the same size, we have

$$\text{Det}(AB) = \text{Det}(A) \text{Det}(B).$$

Example

(Theorem: No proof) Recall that the minors of the matrix A are denoted by $M^{ij}(A)$.

$$\text{Then we have } M^{ij}(A^T) = [M^{ij}(A)]^T.$$

Example

Theorem: The determinant of a matrix and the determinant of its transpose are equal:

$$\text{Det}(A) = \text{Det}(A^T).$$

Theorem: The determinant of a matrix is a linear function of any one column (or row).

Example

Cramer's rule

Theorem: Cramer's rule.

Example

Computing inverses using determinants

Theorem: No proof: The inverse of an $n \times n$ invertible matrix A can be computed by the the formulas

$$(A^{-1})_{ij} = \frac{(-1)^{i+j}}{\text{Det}(A)} \text{Det}(M^{ji}) = \frac{C_{ji}}{\text{Det}(A)}.$$

Example

General exercises 4.2 (no. 1- 26).

7. Vector Subspaces:

Introduction; definition of subspace and example

Theorem: Let U be a subset of a vector space V . Suppose that U contains the zero element and is closed under addition and multiplication by scalars. Then U is a subspace of V .

Theorem: The span of a nonempty set in a vector space is a subspace of that vector space.

Examples

Theorem: The vector sum of two subspaces in a vector space is also a subspace.

Example

Theorem: The intersection of two subspaces in a vector space is also a subspace of that vector space.

But for union does not hold, Example

Theorem : If f is a linear transformation from a vector space X to a vector space Y , then for any subspace U in X , $f[U]$ is a subspace of Y .

Example

Definition of inverse image

Theorem : If f is a linear transformation from a vector space X to a vector space Y , then for any subspace V in Y , $f^{-1}[V]$ is a subspace in X .

Example

Linear transformations

Theorem : The kernel of a linear transformation is subspace in the domain of that transformation.

Examples

Theorem : The range of a linear transformation is a subspace of its co-domain.

Example

Revisiting kernels and null spaces; definition and examples

The row space and column space of a matrix; definition of row space and column space.

Theorem: No proof : If two matrices are row equivalent to each other, then they have the same row space.

Examples

Theorem : The column space of an $m \times n$ matrix A is the set $\{Ax : x \in \mathbb{R}^n\}$

Example

Theorem: No proof : The row space of a matrix A is the column space of A^T .

Examples

General exercises 5.1 (no 1-11, 45)

Basis for a vector space; definition and examples

Theorem : The set of columns of any $n \times n$ invertible matrix is a basis for $\mathbb{R}^n$

Theorem: No proof : If a basis is given for a vector space, then each vector in the space has a unique expression as a linear combination of elements in that basis.

Example

Coordinate vector; definition and example

Theorem : If β is a basis, then the mapping of x to its coordinate vector $[x]_\beta$ is linear.

Isomorphism and equivalence relation; definition of equivalence relation

Theorem : The relation of isomorphism between two vector spaces is an equivalence relation.

Examples

Theorem: No proof : If a vector space has a finite basis, then all of its bases have the same number of elements.

Finite dimensional and infinite dimensional vector spaces; definition and example

Theorem: No proof : Two finite dimensional vector spaces are isomorphic to each other if and only if their dimensions are the same.

Examples

Theorem: No proof : The span of a set is not affected by removing from the set one element that is a linear combination of other elements of that set.

Theorem: No proof : If a set of n vectors spans an n -dimensional vector space, then the set is a basis for that vector space.

Theorem: No proof : In an n -dimensional vector space, every linearly independent set of n vectors is a basis.

Examples

Theorem: No proof : If a set of n vectors spans a vector space, then the dimension of that space is at most n .

Theorem: No proof : If a set of n vectors is linearly independent in a vector space, then the dimension of the space is at least n .

Theorem : The pivot column of a matrix form a basis for its column space.

Examples

Linear transformation of a set; discussion and example

Theorem: No proof : In a finite dimensional vector space, any linearly independent set can be expanded to create a basis.

Dimensions of various subspaces; definition of domain, co-domain, nullity and range and example.

Theorem : If T is a linear transformation whose domain is an n -dimensional vector space, then

$$\text{Dim}(\text{Ker}(T)) + \text{Dim}(\text{Range}(T)) = n.$$

Theorem: No proof : The row space and column space of a matrix have the same dimension:

$$\text{Dim}(\text{Row}(\mathbf{A})) = \text{Dim}(\text{Column}(\mathbf{A})).$$

Examples

General exercises 5.2 (no. 1- 29, 85)

Coordinate vectors; definition and example

Changing coordinates; discussion and examples

Linear transformations; discussion and examples

Theorem : If $\sum_{i=1}^n a_i \mathbf{u}_i = \mathbf{0}$ and if L is linear, then $\sum_{i=1}^n a_i L(\mathbf{u}_i) = \mathbf{0}$

Example

Mapping a vector space into itself; discussion

Similar matrix; definition and examples

More on equivalence relations; discussion and examples

General exercises 5.3 (no. 1, 2, 4, 6, 7, 8, 9, 14, 19, 26, 29, 31)

8. Eigen Systems:

Introduction

Eigenvectors and eigenvalues; definition and example

Using determinants in finding eigenvalues;

Theorem : A scalar λ is an eigenvalue of a matrix $\mathbf{A}$ if and only if $\text{Det}(\mathbf{A} - \lambda \mathbf{I}) = 0$

Examples

Linear transformations; definition and example, definition of diagonalizable and example.

Theorem: No proof : The geometric multiplicity of an eigenvalue cannot exceed its algebraic multiplicity.

Distinct eigenvalues

Theorem : Any set of eigenvector corresponding to distinct eigenvalues of a matrix is linearly independent.

Corollary: No proof : If an $n \times n$ matrix $\mathbf{A}$ has n distinct eigenvalues, then $\mathbf{A}$ is diagonalizable.

Bases of eigenvectors; discussion

Corollary: No proof : If an $n \times n$ real matrix $\mathbf{A}$ has n distinct real eigenvalues, then the corresponding eigenvectors of $\mathbf{A}$ constitute a basis for $\mathbb{R}^n$.

Characteristic equation and characteristic polynomial; definition and example

Diagonalization involving complex numbers; example

Theorem: No proof: If a real matrix A has a complex eigenvalue λ , then the conjugate ($\bar{\lambda}$) is also an eigenvalue. Thus we have $Ax = \lambda x$ and $A\bar{x} = \bar{\lambda}\bar{x}$.

Application: powers of a matrix; discussion and example

General exercises 6.1 (no. 1-21, 27, 46, 49, 50)

9. Inner-Product Vector Spaces:

Inner product spaces and their properties; definition, properties and examples

The norm in an inner product space; definition and examples

Theorem: In any inner-product space, the norm has these properties:

(a) $\|x\| > 0$ for every nonzero vector x

(b) $\|\alpha x\| = |\alpha| \|x\|$

(c) $|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$

(Cauchy-Schwarz Inequality)

(d) $\|x + y\| \leq \|x\| + \|y\|$

(Triangle Inequality)

(e) If $\langle x, y \rangle = 0$, then $\|x + y\|^2 = \|x\|^2 + \|y\|^2$

(Pythagorean Theorem)

(f) $\|x + y\|^2 = \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2$

$\|x + y\|^2 = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2$, in the real case

Distance function; definition and example

Mutually orthogonal vectors; definition and example

Theorem: Every orthogonal set of nonzero vectors in an inner-product space is linearly independent.

Theorem: No proof: In $\mathbb{R}^n$, there does not exist an orthogonal set of $n + 1$ nonzero vectors.

Theorem: No proof: Any orthogonal set of n nonzero vectors in $\mathbb{R}^n$ is a basis $\mathbb{R}^n$.

Example

Orthogonal projection; definition

Theorem: In an inner-product space, the orthogonal projection of a vector x onto a nonzero vector y is the point

$$p = \frac{\langle x, y \rangle}{\langle y, y \rangle} y$$

It has the property that $x - p$ is orthogonal to y . Thus, x is split into an orthogonal pair of vectors in the equation $x = p + (x - p)$.

Example

Angle between vectors; definition and example

Orthogonal complements; definition and example

Theorem: In an inner-product space, the orthogonal complement of a set is the same as the orthogonal complement of its span.

Example

Theorem: In an inner-product space, the orthogonal complement of any subset is a subspace

Example

Theorem: The null space of a matrix is the orthogonal complement of its row space:

$\text{Null}(A) = [\text{Row}(A)]^\perp$ Examples

Orthogonal bases; discussion

Theorem: If $\{u_1, u_2, \dots, u_n\}$ is an orthonormal set in an inner-product space, then each x in the span of that set is expressible as

$x = \sum_{i=1}^n \langle x, u_i \rangle u_i$.

Definition of orthogonal, unitary with examples

Theorem: No proof : Every orthonormal set is linearly independent.

Theorem : $\{u_1, u_2, \dots, u_n\}$ be an orthonormal basis for a subspace U in an inner-product space. The orthogonal projection of any x onto U is the point

$$p = \sum_{i=1}^n \langle x, u_i \rangle u_i$$

Theorem: No proof : In order that a vector be orthogonal to a subspace (in an inner-product space), it is sufficient that the vector be orthogonal to each member of a set that spans the subspace.

Subspaces in inner-product spaces; definition of direct sum and examples

Theorem : If U is a subspace in an n -dimensional inner-product space, then

$$\dim(U) + \dim(U)^\perp = n$$

Theorem : Let U and V be subsets of an inner-product space. If $U \subseteq V$, then $V^\perp \subseteq U^\perp$.

Theorem : If U is a finite-dimensional subspace in an inner-product space X , then X is the direct sum of U and $U^\perp$:

$$X = U + U^\perp,$$

$$U \cap U^\perp = \{0\},$$

$$U = U^{\perp\perp}$$

General exercises 7.1 (no. 1-20, 24, 42, 43,)

The Gram-Schmidt Process; discussion and example

Theorem : Every finite-dimensional inner-product space has an orthonormal basis.

Example

Modified Gram-Schmidt process; discussion and examples

Linear least-square solution; discussion

Theorem: No proof : If a system of equation $Ax = b$ is inconsistent, then the system of normal equation,

$$A^T A x = A^T b$$

can be solved for x , and with this choice of x , the point Ax will be as close to b as possible.

Example

General exercises 7.2 (no. 1-14, 17, 41, 42, 43, 52, 54)

10. Additional Topics:

Introduction; discussion

Hermitian matrices and self-adjoint mapping; discussion, definition and examples

Self-adjoint mapping; definition

Theorem : Every self-adjoint mapping is linear.

Theorem : All eigenvalues of a self-adjoint operator are real.

Theorem : Any two eigenvectors of a self-adjoint operator if associated with different eigenvalues, are mutually orthogonal.

Theorem : Consider $\mathbb{C}^n$ with its standard inner product, and let A be an $n \times n$ matrix. Define a linear transformation L by writing $L(x) = Ax$, where $x \in \mathbb{C}^n$. The operator L is self-adjoint if and only if A is Hermitian.

Corollary: No proof : All eigenvalues of a Hermitian matrix are real.

Corollary: No proof : All eigenvalues of a symmetric real matrix are real.

Theorem : Let L be a linear transformation on a finite-dimensional vector space. If A is the matrix for L relative to a given basis, then the eigenvalues of L and A are the same.

Examples

Unitary and orthogonal matrices; definition and examples

The Cayley- Hamilton Theorem:

Cayley- Hamilton Theorem: Every square matrix satisfies its own characteristic equation.

tion.

Examples upto 3×3 matrix.

Quadratic forms; discussion definition and example

Theorem : If $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ real matrices connected by the relation

$$\mathbf{B} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^T)$$

then the corresponding quadratic forms of $\mathbf{A}$ and $\mathbf{B}$ are identical, and $\mathbf{B}$ is symmetric.

Examples

General Exercises 8.1 (no. 1- 17, 26)

Permutation matrix; definition and examples

LU-factorization; discussion and examples

QR-factorization; discussion and examples

Theorem: No proof : The vector $\mathbf{x}$ is the least-squares solution of the system of equations $\mathbf{Ax} = \mathbf{b}$ under the circumstances just described.

Theorem: No proof : Any $m \times n$ matrix of rank n has a QR- factorization, where $\mathbf{Q}$ is an $m \times n$ matrix, $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}_n$, and $\mathbf{R}$ is an $n \times n$, upper triangular, and invertible matrix.

Example

Partitioned matrices; definition, discussion, and examples

Solving a system having a 2×2 block matrix; discussion

General exercises 8.2 (no. 1-17, 33 37, 40)

Richardson iterative method; discussion and example

Jacobi iterative method; discussion and example

Gauss-Seidel method; discussion and example

Guidelines For Question Setter

The students will not have to opportunity to leave some units and equal importance has to be given to each unit. We would like to draw attention which might be helpful to the teachers and the question setters/ examiners. The main point to be emphasized is that the concept part must constitute an integral part of the question, but it does not mean that the problems, applications, and accuracy are less important. So the related problems are to be incorporated as a part or as a whole as the case may be.

There will be 10 questions divided in two groups A and B.

All the questions are compulsory.

Group A has 5 questions and each carry 8 marks and there will be two OR (choice) in same unit only in any question number.

Group B has also 5 questions and each carry 7 marks and there will be two OR (choice) in same unit.

On the basis of the guidelines mentioned, we enclose a set of model questions for MAT 201 (Linear Algebra).

Model Question
Tribhuvan University
Institute of Science and Technology

Bachelor Level(4 Yrs.) Science and Technology/ II year
Linear Algebra
Mathematics (MAT 201)

Full Marks:75
Pass Marks: 27
Time: 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt ALL questions.

Group A

$5 \times 8 = 40$

1. When the system of equation is consistent and inconsistent? Write three forms of solution. Solve the system of equations by Gauss elimination method $3x_1 + 6x_2 + 6x_3 = 21$, $2x_1 + 64x_2 + 5x_3 = 16$ and $2x_1 + 5x_2 + 4x_3 = 17$. [2+1+5]

2. In $\mathbb{R}^2$, if a triangle has vertices $\mathbf{0}$, $\mathbf{u}$, and $\mathbf{v}$, then prove that the area of the triangle is one-half of the absolute value of the 2×2 determinant having rows (or columns) $\mathbf{u}$ and $\mathbf{v}$:

$$\text{Area}[\Delta(\mathbf{0}, \mathbf{u}, \mathbf{v})] = \frac{1}{2} \left| \text{Det} \begin{bmatrix} u_1 & u_2 \\ v_1 & v_2 \end{bmatrix} \right|$$
, where the notation $\Delta(\mathbf{0}, \mathbf{u}, \mathbf{v})$ specifies a triangle having vertices at $\mathbf{0}$, $\mathbf{u}$, and $\mathbf{v}$.

Use this to find the area of Δ whose vertices are $(4,7)$, $(-2,11)$ and $(12,-6)$. [6+2]

OR

For square matrices of the same size, prove that $\text{Det}(\mathbf{AB}) = \text{Det}(\mathbf{A}) \text{Det}(\mathbf{B})$ and verify

this for the pair of matrices $\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 4 \\ 1 & 5 & 1 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 2 & 2 & 3 \\ -1 & 1 & 3 \\ 5 & 1 & 4 \end{bmatrix}$. [5+3]

3. Define an eigenvalue of a matrix. Prove that a scalar λ is an eigenvalue of a matrix $\mathbf{A}$ if and only if $\text{Det}(\mathbf{A} - \lambda\mathbf{I}) = 0$. What are the characteristic equation, the eigenvalues and the eigenvectors of the matrix $\begin{bmatrix} 2 & 1 \\ 4 & -1 \end{bmatrix}$? [1+ 2 +5]

4. Define a linear mapping. Let $\mathbf{A}$ be an $m \times n$ matrix. Prove that The mapping $\mathbf{x} \mapsto \mathbf{Ax}$ is linear from $\mathbb{R}^n$ to $\mathbb{R}^m$. Also, if a mapping f is defined from $\mathbb{R}^2$ to $\mathbb{R}^3$ by the formula

$$f(s, t) = f\left(\begin{bmatrix} s \\ t \end{bmatrix}\right) = \begin{bmatrix} 2s + 3t \\ -s + 5t \\ 4s - 3t \end{bmatrix}$$
. Is it a linear map? Justify your answer. [1+3+4]

5. Define an inner product space. Prove that $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|$ and verify this when $\mathbf{x} = (-1, 0, 1)$ and $\mathbf{y} = (0, 2, -1)$. [1+4+3]

OR

Write the formula of Gram-Schmidt process and use it to find the orthonormal basis of the vectors $\mathbf{v}_1 = (1, 0, 3)$, $\mathbf{v}_2 = (2, 2, 0)$, and $\mathbf{v}_3 = (3, 1, 2)$ [2+6]

Group B

$$5 \times 7 = 35$$

6. Find a matrix whose kernel is spanned by the two vectors $\mathbf{u} = (1, 3, 2)$ and $\mathbf{v} = (-2, 0, 4)$ [7]

Ward Cheney Ex 1.3 QN 6.

7. The expression $(12, 28) + t(9, 21)$ describes a line in $\mathbb{R}^2$ when t turns over all of $\mathbb{R}$. Does this line pass through origin? Justify your answer. [7]

OR

Define a composite map. If S and T are linear maps, then show that $S \circ T$ is also linear. [1+6]

8. Find a linear transformation that maps $(1, 3, -7)$ to $(2, 0, 2)$ and $(3, 2, 1)$ to $(-1, 1, 5)$ and $(-3, 5, -23)$ to $(8, -2, 5)$. [7]

(Ex-6 Page 348) (Ward Cheney)

OR

If $\sum_{i=1}^n a_i \mathbf{u}_i = \mathbf{0}$ and if L is linear, then $\sum_{i=1}^n a_i L(\mathbf{u}_i) = \mathbf{0}$ [7]

9. Define an invertible matrix. Here are two matrices $\mathbf{A} = \begin{bmatrix} 3 & 7 & 5 \\ 5 & 11 & 8 \\ 3 & 4 & 3 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 1 & -1 & 1 \\ 9 & -6 & 1 \\ -13 & 9 & -2 \end{bmatrix}$. Is one of these the inverse of the other? If so, which is which?

[1+6]

10. Find the LU-factorization of the matrix $\begin{bmatrix} 2 & -4 \\ 10 & -17 \end{bmatrix}$. [7]